

'Made For Mobile' Microprocessors

There is a great degree of mobile-desktop convergence happening, as palpable from the current-generation tablets and smartphones. Yet, mobile devices have special requirements that require special processors

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"Mobile processors are designed with the approach of doing more with less. This is one of the reasons why you do not see x86-based smartphones today," quips James Bruce, lead mobile strategist at ARM. In the case of desktop processors, power efficiency, size of the chip and thermal management are not limiting factors, though these are important design aspects. However, in the case of a mobile device, which needs to compete with a desktop in terms of functionality and at the same fit into your pocket, working for long hours on battery, these 'mobility' factors are critical.

Bruce explains that desktop processors are 'processors on chip.' Many surrounding chips are needed to complete the compute sub-system. On the other hand, in the mobile world, the complete sub-system is on a single chip—a system-on-chip (SoC).

"For example, you may have a 3G

(or even 4G) modem, processor, graphics, video unit, audio decoder, Wi-Fi, Bluetooth, geographical positioning system (GPS), dynamic random-access memory (DRAM) and Flash, all in one 14×14mm² package," Bruce says.

Praful Joshi, business development manager, WindRiver India, concurs, "The key challenge to an engineer designing a mobile processor is how to add more functionality while designing an SoC without compromising on low power consumption or low form factor. In fact, the processor chip size is a very important factor. You can see new very-large-scale integration (VLSI) technologies such as 30 nm or less being widely used in mobile processors as against desktop processors."

"Today, consumers expect to have the same kind of computing experience on their mobile device as on their desktop or notebook PC—high-definition (HD) video playback, audio and video streaming, multitasking, Web browsing, 3D gaming and 3D interfaces. However, the power envelopes of a PC and a mobile device are radically different. Simply miniaturising a traditional x86 CPU, which you would find in vast majority of PCs, is not enough. The resulting processor might physically fit into a mobile device but its power consumption would result in a device with a battery life of

minutes," says Vishal Dhupar, managing director, Asia-South, NVIDIA.

Dhupar recalls that when creating NVIDIA's mobile solution Tegra, their engineers had started from the ground up with a strict power budget. The designers had to fight for every milliwatt of power. All of this shows that a mobile being different from a desktop, its processor also needs to be different.

War of the titans

Quite naturally, the repertoire of made-for-mobile microprocessors is also increasing, with many ARM-based offerings from Texas Instruments (TI), Qualcomm, NVIDIA, MediaTek, etc. and Intel Atom chips also catching up now.

"ARM is the most popular processor in the mobile device world. It is not a derivative of any desktop processor. It was designed specifically for low-power-consuming devices. Since then it has emerged as a mainstream mobile processor and added more computational capabilities and a software ecosystem. ARM licenses its core processors to SoC vendors such as TI, Qualcomm and others, who, in turn, add their own unique IP for different mobile markets," says Joshi of WindRiver—an Open Source operating systems (Android, Linux and Tizen) commercialisation partner for mobile processor designers as well as mobile device manufacturers.

ARM has several processors for designers to choose from, including the latest Cortex-A9, which features 2GHz typical operation with the TSMC 40G hard macro implementation, low-power targeted single-core implementations into cost-sensitive devices,

